THE UNIVERSITY OF AZAD JAMMU AND KASHMIR

MUZAFFARABAD

DEPARTMENT OF SOFTWARE ENGINEERING

LAB 1

Course: Software Construction & Development (Lab)

Semester: 5th

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Question 1: Online Ticket Reservation System

Task: Develop a use case diagram that can serve as a basis for understanding the requirements of an online ticket reservation system.

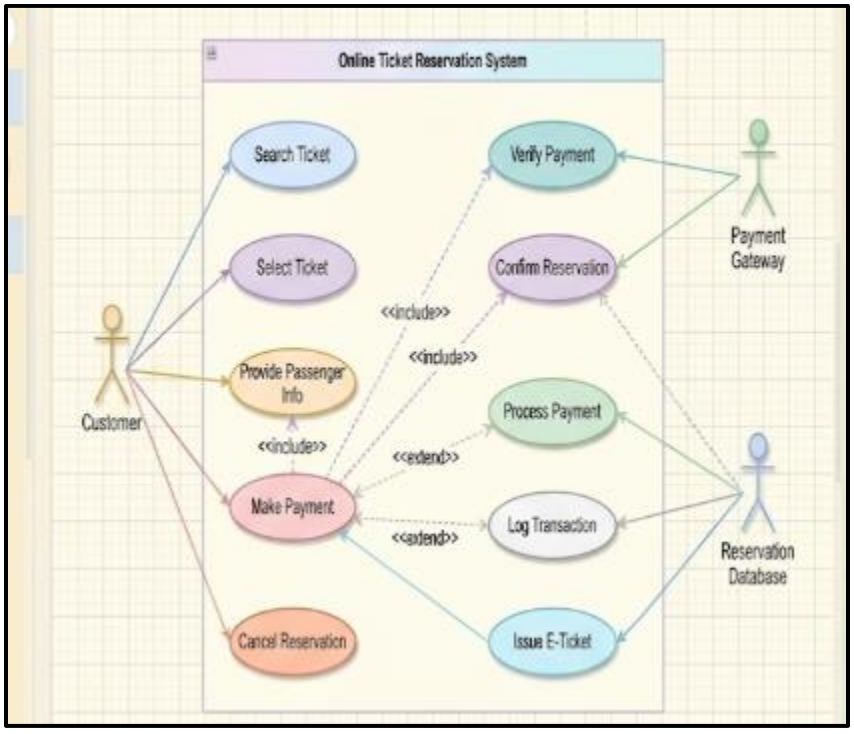
Actors

• Customer: Searches for tickets, checks availability, selects a ticket, enters passenger details, makes payment, confirms a reservation, receives an e-ticket, and may cancel a reservation.

Use Cases

- Search Ticket — allows the customer to search for available journeys or tickets.
- Check Availability — shows whether a suitable ticket is available.
- Select Ticket — allows the customer to choose a ticket.
- Enter Passenger Details — records the passenger information required for booking.
- Make Payment — allows the customer to pay for the reservation.
- Confirm Reservation — confirms the booking after the required steps are completed.
- Receive E-Ticket — provides the customer with the electronic ticket.
- Cancel Reservation — allows the customer to request cancellation of a reservation.

Use Case Diagram



Note: The diagram represents a reasonable proposed design because the lab question does not specify a detailed booking workflow or particular payment rules.

Question 2: Credit Card Processing System

Task: Identify the actors and use cases in the given case study, then draw a use case diagram.

Actors

• Customer: Selects an item, provides the credit card for payment, and signs the receipt.

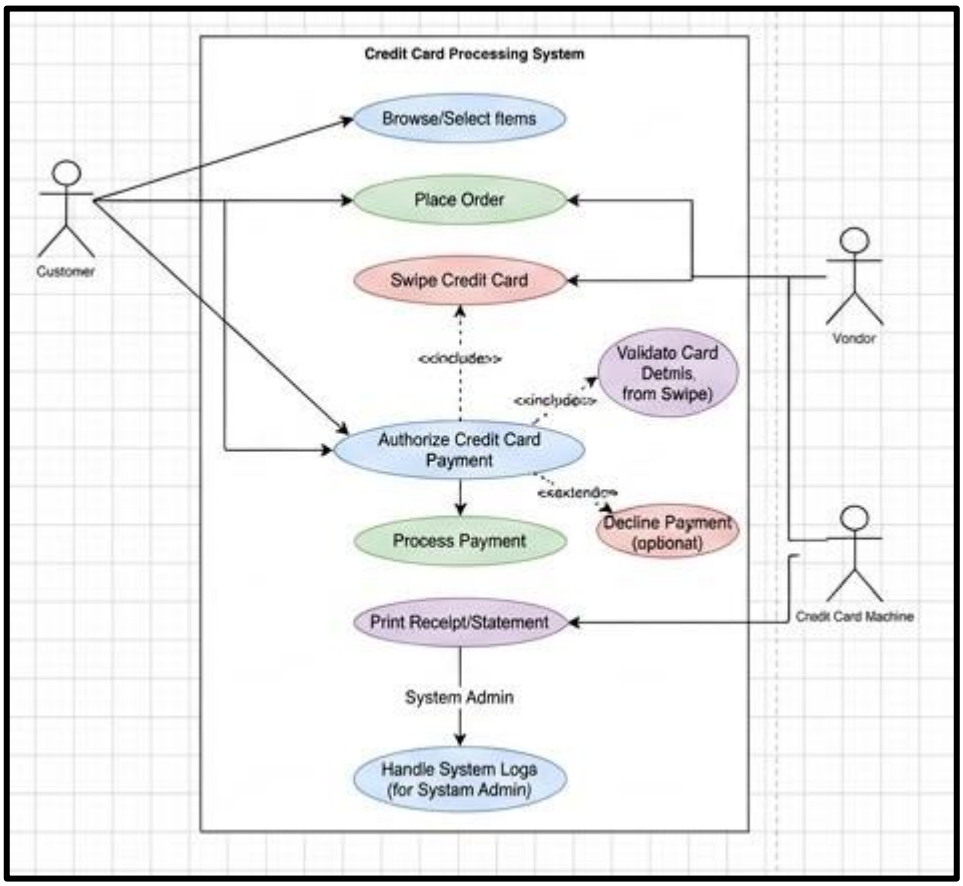
• Vendor: Prepares the bill, handles the card-swiping step, issues the receipt or balance statement, and receives the signed receipt.

• Credit Card Machine: Processes the requested transaction and prints the receipt or balance statement. This actor represents the machine described in the case study.

Use Cases

- Select Item — the customer selects the item to purchase.
- Generate Bill — the vendor prepares a bill for the selected item.
- Swipe Credit Card — the customer's card is swiped using the card machine.
- Process Transaction — the card machine processes the requested amount transaction.
- Print Receipt / Balance Statement — the machine provides a printed transaction receipt or balance statement.
- Sign Receipt — the customer signs the receipt.
- Return Signed Receipt — the customer returns the signed receipt to the vendor.

Use Case Diagram



Conclusion

Use case diagrams help identify the people or external systems that interact with a system and the main services the system provides. The first diagram describes the major customer activities in online ticket reservation. The second diagram represents the customer, vendor, and credit card machine interactions during a shop purchase.

*****THE END*****